

GATE SOLUTION IN MINING ENGINEERING

(Containing Solved Questions 2015 - 2007)

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Gate Solution -2009

Q.1 If A is an orthogonal matrix, then

- (A) $A^T = A^{-1}$ (B) $A^T = -A^{-1}$ (C) $A = A^{-1}$ (D) $A = -A^{-1}$

Answer (A)

Solution .

In case of the orthogonal matrix –

$$A A^T = A^T A = I$$

Or $A^T = A^{-1}$

Q.2 In a normal (Gaussian) distribution curve, the area between one standard deviation from mean on either side in percent is

- (A) 50 (B) 68 (C) 86 (D) 95

Answer (B)

Q3. A measure of dispersion of a sample data set is

- (A) mean (B) median (C) mode (D) standard deviation

Answer (D)

Q 4. The value of limit $\lim_{x \rightarrow 2} \left(\frac{2\sqrt{4-x^2}}{5} \right)$ is

- (A) $\frac{-2\sqrt{8}}{5}$ (B) 0 (C) $\frac{2\sqrt{8}}{5}$ (D) Non-existent

Answer (B)

Solution .

$$\begin{aligned} & \lim_{x \rightarrow 2} \left[\frac{2\sqrt{4-x^2}}{5} \right] \\ & \rightarrow \left[\frac{2\sqrt{4-2^2}}{5} \right] \end{aligned}$$

→ 0

Q 5. $\hat{i}, \hat{j}$ and $\hat{k}$ represent the unit vectors in the positive x, y and z directions of a Cartesian coordinate system. Using the right- hand rule, $\hat{k} \times \hat{j}$ represents

- (A) 0 (B) 1 (C) $-\hat{i}$ (D) $\hat{i}$

Answer (C)

Q 6. The rock mass classification system that considers "active stress" factor is
(A) Q-system (B) RMR (C) RQD (D) GSI

Answer (A)

Q 7. In a triaxial compression test if σ_1 is a axial stress and σ_2 and σ_3 are confining stresses, then
(A) $\sigma_3 > \sigma_2 = \sigma_1$ (B) $\sigma_1 > \sigma_2 = \sigma_3$ (C) $\sigma_1 = \sigma_2 > \sigma_3$ (D) $\sigma_3 = \sigma_2 > \sigma_1$

Answer (B)

Q 8. In a longwall mining subsidence phenomenon, the "angle of break" is the angle between

- (A) the vertical line at the panel edge and line connecting the panel edge and zero subsidence on the surface
(B) the vertical line at the panel edge and line connecting the panel edge and point of critical deformation on the surface
(C) the vertical line at the panel edge and line connecting the panel edge and the point of the maximum tensile strain on the surface
(D) the horizontal line and the line connecting the panel edge and zero subsidence on the surface

Answer (C).

Q 9. Pocket and wing technique of pillar extraction is related to

- (A) bord and pillar method (B) room and pillar
(C) Wongawilli method (D) shortwall method

Answer (A)

Q 10. A non-electric detonating relay DOES NOT contain

- (A) delay element (B) fuse head
(C) metal sleeve (D) neoprene connecting tube

Answer (C)

Q 11. An iron ore deposit has a mean grade of 63% Fe. During the course of mining, 30% fines by weight are generated at a grade of 72% Fe which are rejected. The effective mean grade of the deposit in FE percentage is

- (A) 59.1 (B) 53.1 (C) 50.4 (D) 41.4

Answer (A)

Solution .

Assume W is insitu tonnage of Fe deposit –

$$\begin{aligned}\therefore \text{effective mean grade. (Fe \%)} &= \frac{(0.63 \times W - 0.3 \times 0.72 \times W)}{0.7 \times W} \times 100 \\ &= \frac{0.414}{0.7} \times 100 \\ &= 59.1\%\end{aligned}$$

Q 12. Koepe system of winding does NOT include

- (A) taper guide (B) limit switches
(C) safety hook (D) brake

Answer (C)

- Q 13. A gas mask does NOT include
 (A) check valve (B) warning device
 (C) face piece assembly (D) coolant canister

Answer (D)

- Q 14. Resuing stoping method is adopted when ore body is
 (A) flat and thick (B) very steep and thick
 (C) flat and thin (D) very steep and thin

Answer (A)

15. Moody diagram represents resistance coefficient in terms of
 (A) Reynolds number and asperity ratio (B) viscosity and aspect ratio
 (C) surface tension and viscosity (D) Reynolds number and surface tension

Answer (A)

- Q 16. An area of 100 m^2 is measured on a plan having R.F. of $1/800$. If the R.F. were to be $1/2000$,
 The area in m^2 would be
 (A) 16 (B) 40 (C) 250 (D) 625

Answer (D).

Solution.

$$\text{R.F.} = \frac{\text{Drawing demension of object}}{\text{actual dimension of object}}$$

Given, actual area of object = 100 m^2

Actual length or width of object = $\sqrt{100} = 10 \text{ m}$

$$\frac{1}{800} = \frac{\text{Drawing length or width of object}}{10}$$

$\therefore$ Drawing length or width of object = 0.0125 m

Now,

$$\frac{1}{2000} = \frac{0.0125}{\text{actual length or width of object}}$$

∴ actual length or width of object = 25 m

∴ actual area of object = 625 m²

- Q 17.** As per the DGMS norms, the severity index is a measure of
(A) fatality rate (B) serious injury rate
(C) number of reportable injuries (D) accident proneness of mine

Answer (D)

- Q 18.** A balanced transportation problem is characterized by

- (A) total supply exceeds total demand
(B) total demand exceeds total supply
(C) total demand is equal to total supply
(D) total supply is either equal to or more than total demand

Answer (C)

- Q 19.** In the context of project management techniques, the TRUE statement is

- (A) CPM is stochastic and PERT is deterministic
(B) CPM is deterministic and PERT is stochastic
(C) Both CPM and PERT are deterministic
(D) Both CPM and PERT are stochastic

Answer (B)

- Q20.** For mining property appraisals, typical reports prepared are Bankable Feasibility report (BFR), Conceptual Plan Report (CPR), Feasibility Report (FR) and Detailed Project Report (DPR). The

chronological order for the preparation of these reports is

- (A) CPR → FR → BFR → DPR (B) BFR → CPR → DPR → FR
(C) FR → BFR → CPR → DPR (D) CPR → BFR → DPR → FR

Answer (C)

Q 21. The mean of the cubes of the first n natural numbers is

- (A) $\frac{n(n+1)^2}{4}$ (B) $\frac{n(n+1)(n+2)}{8}$ (C) $\frac{n^4+1}{n}$ (D) $\frac{n^3}{4}$

Answer (A)

Q 22. The sum of the eigenvalues of the matrix $\begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$ is

- (A) -3 (B) -1 (C) 1 (D) 3

Answer (C)

Solution .

The sum of the Eigen value of a matrix is the sum of the element of the principal diagonal.

$$\therefore 1+0 = 1$$

Q 23. The value of $\nabla \cdot F$ of a vector $F = 4x^2\hat{i} + 3xy^2\hat{j} + xyz^3\hat{k}$ at the point (1, 1, 2) is
(A) 24 (B) 26 (C) 30 (D) 32

Answer (B)

Solution .

$$\vec{F} = 4x^2\hat{i} + 3xy^2\hat{j} + xyz^3\hat{k}$$

$$\nabla \cdot F = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (4x^2\hat{i} + 3xy^2\hat{j} + xyz^3\hat{k})$$

$$\nabla \cdot F = (8x + 6xy + 3xyz^2)$$

At point (1,1,2)

$$\nabla \cdot F = (8 \times 1 + 6 \times 1 \times 1 + 3 \times 1 \times 1 \times 4)$$

$$\nabla \cdot F = 26$$

Q 24. The function $f(x) = x^3(1-x)$ is integrated between 0 and 1 (both inclusive) using closed form method and also by Simpson's 1/3 rule. the difference in the values obtained from

these methods is

- (A) 0 (B) 1/480 (C) 1/120 (D) 1/20

Answer (C)

Solution .

→ closed form method

$$\int_0^1 f(x)dx = \int_0^1 x^3(1-x)dx$$

$$\int_0^1 f(x)dx = \left| \frac{x^4}{4} - \frac{x^5}{5} \right|_0^1$$

$$\therefore \int_0^1 f(x)dx = \frac{1}{20}$$

→ Simpson's 1/3 rule

$$\int_a^b f(x)dx = \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

$$\int_0^1 x^3(1-x)dx = \frac{1-0}{6} \left[0 + 4 \times \frac{1}{8 \times 2} + 0 \right]$$

$$\int_0^1 x^3(1-x)dx = \frac{1}{24}$$

Therefore,

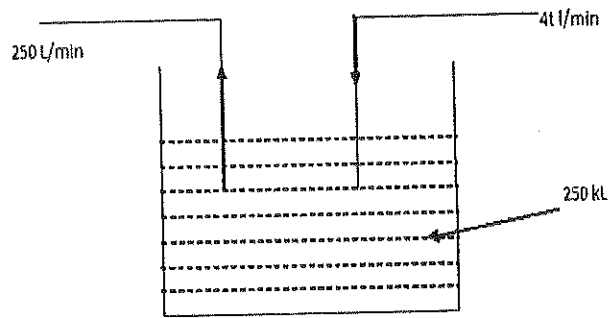
$$\text{Difference in values} = \frac{1}{20} - \frac{1}{24} = \frac{1}{120}$$

- Q 25. Water starts to flow into a sump initially containing 250 kL of water. The inflow rate of water is $4t$ L/min where t refers to time elapsed in min. If the pumping rate of water out of the sump is 250 L/min, the total volume of water in the sump after 3 hours in KL is

- (A) 250.5 (B) 255.6 (C) 269.8 (D) 280.9

Answer (C)

Solution .



In 3 hours total water that has discharged from sump -

$$D_s = \frac{250 \times 3 \times 60}{1000} = 45 \text{ KL}$$

In 3 hours total water inlet into the sump -

$$I_s = \int_0^{180} 4t \, dt = 2t^2 \Big|_0^{180}$$

$$I_s = 64.8 \text{ KL}$$

∴ Total water in sump after 3 hours -

$$W_s = 250 + 64.8 - 45$$

$$W_s = 269.8 \text{ KL}$$

Q 26. There are 50 lemon trees in a reclaimed mine area. Each tree produces 800 lemons per year. For each additional tree planted in this area, considering all trees, the output number of fruits per tree drops by 10 lemons in a year. The number of trees that to be added to the existing reclaimed area in order to maximize the total number of lemons in the year is

- (A) 10 (B) 15 (C) 16 (D) 26

Answer (B)

Solution :

Before planted additional tree , total number of lemon production = $50 \times 800 = 40000$

Let x number of trees added to maximize number of lemons.

Then, total number of tree = $(50+x)$

But output number of lemons drops by 10 per tree in a year.

So, total number of lemons per tree in a year = $(800 - 10x)$

Therefore, total production $(P) = (50 + x) \cdot (800 - 10x)$

$$\text{Now, } \frac{dP}{dx} = (50 + x)(-10) + (800 - 10x) \times 1$$

$$\text{For maximize, } \frac{dP}{dx} = 0$$

$$-500 - 10x + 800 - 10x = 0$$

$$-20x = -300$$

$$x = 15$$

Q 27. The grain density and bulk density of a dry coarse grained sandstone rock sample are 3.0 gm/cc and 2.7 gm/cc respectively. The void ratio of the sample in percentage is

- (A) 8.4 (B) 10.0 (C) 11.1 (D) 30.5

Answer (C)

Solution .

We know,

$$\text{Void ratio} = \frac{\text{volume of void}}{\text{volume of grain}} = \frac{\text{volume of sample} - \text{volume of grain}}{\text{volume of grain}}$$

$$\text{Void ratio} = \frac{\text{volume of sample}}{\text{volume of grain}} - 1$$

$$\text{Void ratio} = \frac{\frac{\text{mass of sample}}{\text{density of sample}}}{\frac{\text{mass of grain}}{\text{density of grain}}} - 1$$

$$\text{Void ratio} = \frac{\frac{\text{mass of grain}}{\text{density of sample}}}{\frac{\text{mass of grain}}{\text{density of grain}}} - 1$$

$$\text{Void ratio} = \frac{\text{density of grain}}{\text{density of sample}} - 1$$

$$\text{Void ratio} = \left(\frac{3}{2.7} - 1 \right) = 0.111 = 11.1\%$$

- Q 28.** The ratio of uniaxial compressive strength to uniaxial tensile strength of a sandstone specimen is 8:1. The theoretical value of angle of internal friction of the specimen in degree is
 (A) 51 (B) 41 (C) 32 (D) 7
 Answer (A)

Solution .

We have,

$$\frac{\sigma_c}{\sigma_t} = \frac{1 + \sin \phi}{1 - \cos \phi}$$

$$\therefore \frac{1}{8} = \frac{1 + \sin \phi}{1 - \cos \phi}$$

$$\therefore \sin \phi = \frac{7}{9}$$

$$\therefore \phi = 51^\circ$$

- Q 29.** A circular tunnel is made underground where far field vertical and horizontal stresses are P_0 and KP_0 respectively. The tangential stress ($\sigma_{\theta\theta}$) at the boundary of the tunnel for $\theta = 45^\circ$ from the horizontal plane is $3P_0$. The value of K is

(A) 0 (B) 1 (C) 2 (D) 3

Answer (C)

Solution .

$$\text{Tangential stress } (\sigma_{\theta\theta}) = P_z [(1 + k) - 2(1 - k) \cos 2\theta]$$

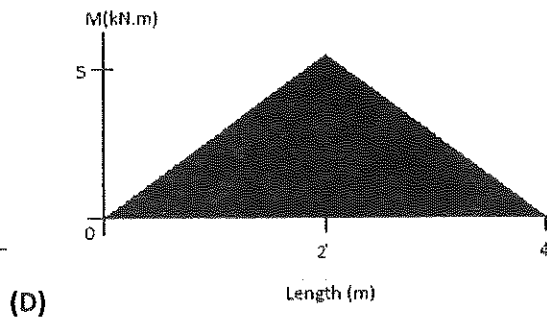
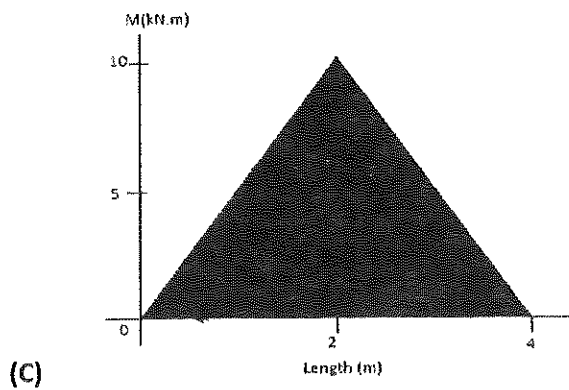
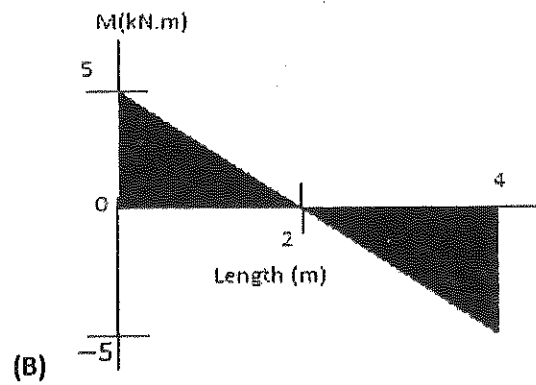
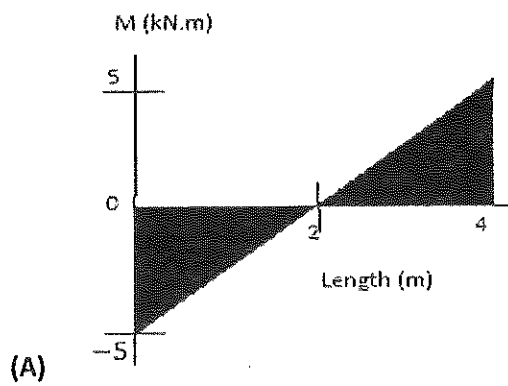
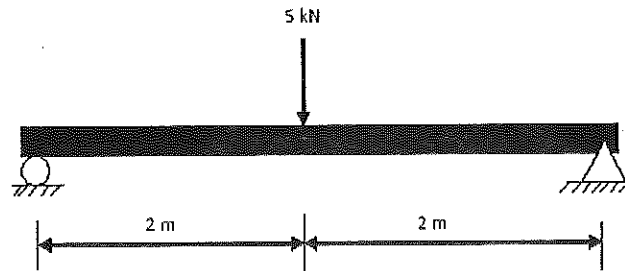
$$\text{Given that, vertical stress } (P_z) = P_0$$

$$\text{And tangential stress } (\sigma_{\theta\theta}) = 3P_0, \theta = 45^\circ$$

$$\therefore 3P_0 = P_0 [(1 + k) - 2(1 - k) \cos(2 \times 45^\circ)]$$

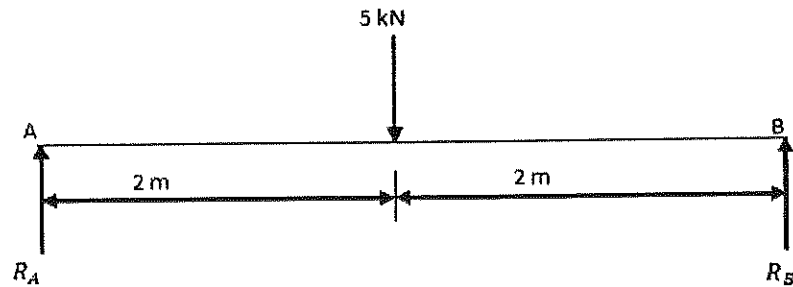
$$\therefore K = 2$$

Q 30. The bending moment diagram for the shaft shown below resembles which one of the following graphs?



Answer (D)

Solution .



$$\therefore \sum M_B = 0$$

$$R_A \times 4 - 5 \times 2 = 0$$

$$R_A = 2.5 \text{ KN}$$

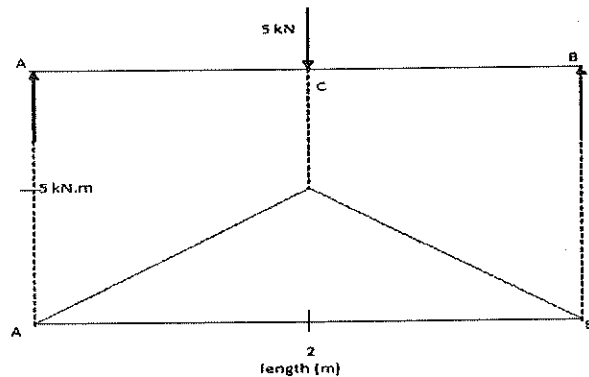
$$\text{And also, } \sum V = 0$$

$$R_A + R_B - 5 = 0$$

$$\therefore R_B = 2.5 \text{ KN}$$

Now, bending moment calculation

$$M_A = 0, \quad M_B = 0, \quad M_C = 2.5 \times 2 = 5 \text{ KN.m}$$



Q 31. A mining equipment has a life of 5 years with no salvage value. Assuming that the depreciation of the equipment is calculated by the straight line method, the average annual value of equipment in percentage of its original value is

- (A) 20 (B) 40 (C) 50 (D) 60p

Answer (A)

Solution .

We have ,

$$\text{Annual depreciation cost} = \frac{\text{first cost} - \text{salvage value}}{\text{life in years}}$$

Given that, life in years = 5

Solage value = 0.

$$\text{The average annual value (D) of equipment in percentage} = \frac{\text{first cost (original cost)} - 0}{5} \times 100$$

D = 20 % of first cost (original cost)

Q32. Air flows at $2 \text{ m}^3/\text{s}$ through a forcing fan duct of 0.3 m^2 having uniform cross-section. The Duct resistance is $40 \text{ N s}^2 \text{ m}^{-8}$ and air density is 1.2 kg/m^3 . The total pressure generated by the fan is Pa is

(A) 186.7 (B) 160.0 (C) 133.3 (D) 26.7

Answer (A)

Solution .

Total pressure (P_T) generated by the fan = velocity pressure + static pressure

$$P_T = \frac{1}{2} \times v^2 \times \rho + R \times Q^2$$

$$P_T = \frac{1}{2} \times \rho \times \left(\frac{Q}{A}\right)^2 + R \times Q^2$$

$$\therefore P_T = \frac{1}{2} \times 1.2 \times \left(\frac{2}{0.3}\right)^2 + 40 \times 2^2$$

$$\therefore P_T = 186.7 \text{ Pa}$$

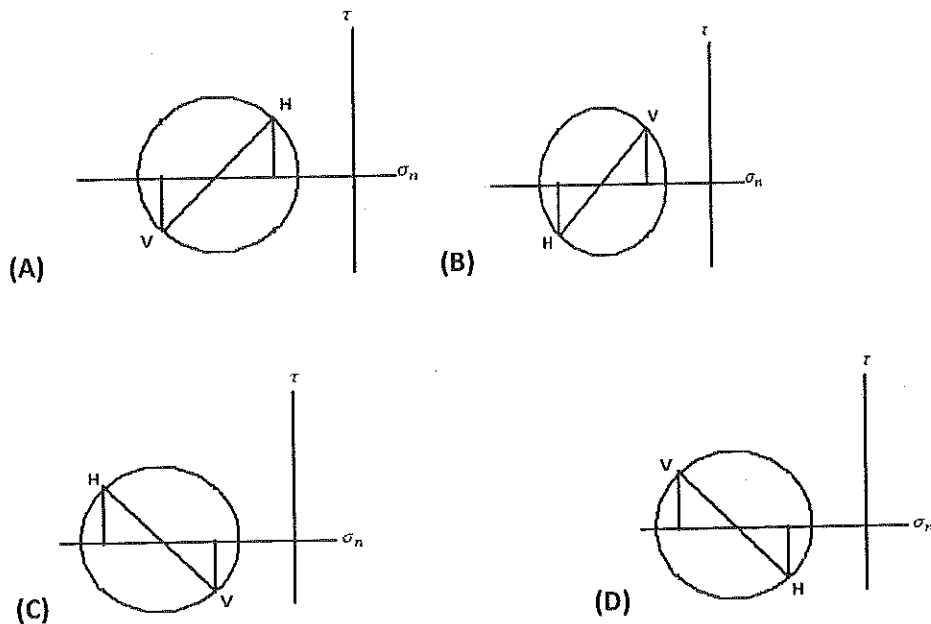
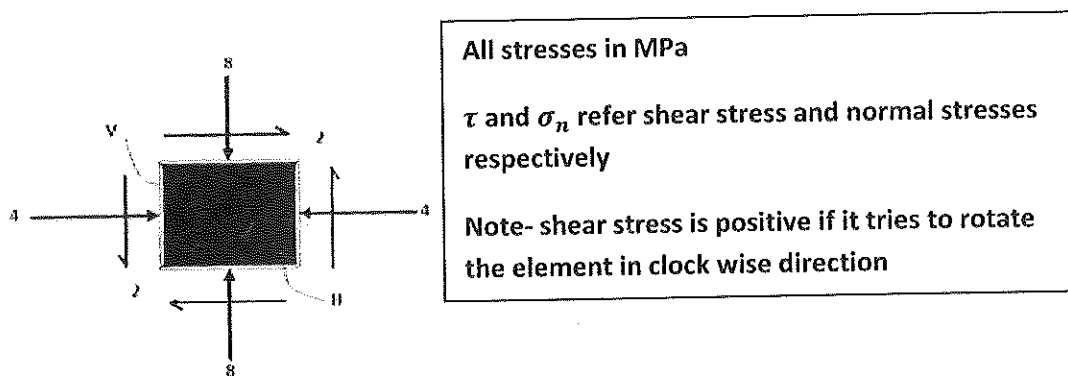
Q 33. Match the following in the context of Indian mining practice:

Equipment	Power source
P. Rocker shovel	1. Battery
Q. Locomotive	2. Compressed air
R. Shearer	3. Electricity (maximum voltage 6.6 kV AC)

- S. Dragline (24 m^3 bucket capacity) 4. Electricity (maximum voltage 1.1 kV AC)
- (A) P - 1, Q - 2, R - 3, S - 4 (B) P - 2, Q - 1, R - 4, S - 3
- (C) P - 2, Q - 1, R - 3, S - 4 (D) P - 1, Q - 3, R - 2, S - 4

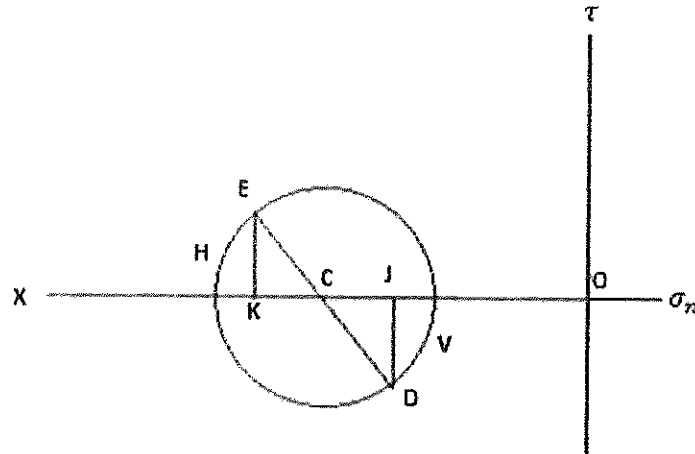
Answer (B)

Q 34. The planes H and V represent the horizontal and vertical planes respectively as shown in the figure. Which one of the following Mohr circles represents the stress conditions applied in planes H and V?



Answer (C)

Solution .



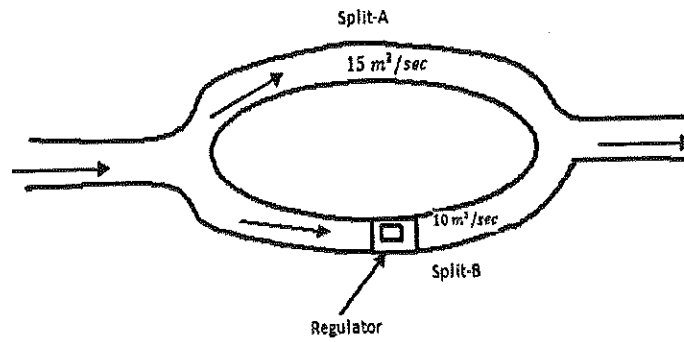
Note –

1. Take point O and draw a horizontal line OX, cut off OJ and OK equal to (-4 MPa) and (-8 MPa) respectively.
2. Now erect a perpendicular at J below line OX and cut off JD equal to negative shear stress (-2 MPa) .
3. Similarly erect a perpendicular at K above OX line and cut off KE equal to positive shear stress (2 MPa) . join DE and bisect it at C.
4. Now with C as centre and radius equal to CD or CE, draw Mohr circle of stresses.

Q 35. Two splits A and B are ventilated from an intake airway. Resistances of the splits are $0.5 \text{ N s}^2 \text{ m}^{-8}$ and $0.8 \text{ N s}^2 \text{ m}^{-8}$ respectively. A regulator is placed in split B to maintain a flow of $15 \text{ m}^3/\text{s}$ and $10 \text{ m}^3/\text{s}$ in splits A and B respectively, as shown in the figure. The size of the regulator in m^2 is

- (A) 2.10 (B) 1.30 (C) 1.20 (D) 1.13

Answer (A).



5.

Solution .

For split A

$$P_A = R_A \times Q_A^2$$

$$P_A = 0.5 \times 15^2 = 112.5 \text{ Pa}$$

For split B

$$P_B = (R_B + R) \times Q_B^2$$

$$P_B = (0.8 + R) \times 10^2$$

Where R-resistance of regulator

$$\text{Here, } P_A = P_B$$

$$112.5 = (0.8 + R) \times 10^2$$

$$\therefore R = \frac{112.5}{100} - 0.8$$

$$R = 0.325 \text{ N s}^2 \text{ m}^{-8}$$

And we know,

$$\text{Size of regulator (A)} = \frac{1.2}{\sqrt{R}}$$

$$A = \frac{1.2}{\sqrt{R}}$$

$$A = \frac{1.2}{\sqrt{0.325}} = 2.10 \text{ m}^2$$

Q 36. The concentration of OH^- ion in a mine water sample is 10^{-11} mol/L . the pH of sample is

- (A) 2 (B) 3 (C) 4 (D) 11

Answer (B).

Solution .

We have,

$$p(OH^-) = -\log[OH^-]$$

$$p(OH^-) = -\log[10^{-11}]$$

$$\therefore p(OH^-) = 11$$

$$\text{Since } p(H^+) + p(OH^-) = 14$$

$$\therefore p(H^+) + 11 = 14$$

$$\therefore p(H^+) = 3$$

Q 37. A mine having a reserve of 320 Mt produces 4 Mt of ore at the end of 1st year. If the mine increase production by 10% every year, the percentage of the reserve that still remains at the end of 21st year is

- (A) 50 (B) 35 (C) 25 (D) 20

Answer (D)

Solution .

Percentage of reserve (R) that still remains at the end of 21th year:-

$$R = \frac{320 - 4\{1 + (1 + 0.1) + (1 + 0.1)^2 + \dots + (1 + 0.1)^{20}\}}{320} \times 100$$

$$R = \frac{320 - 4\{(1.1)^0 + (1.1)^1 + (1.1)^2 + \dots + (1.1)^{20}\}}{320} \times 100$$

$$R = \frac{320 - 4\left\{\frac{1-(1.1)^{21}}{1-1.1}\right\}}{320} \times 100$$

$$R = 20\%$$

Q38. Match the following:

Type of ore deposit	ore, rock strength	Mining method
P. flat, thin	1. Strong, strong	a. Sublevel stoping
Q. Massive	2. Weak, weak	b. room and pillar
R. Steep, thick		c. Block caving
(A) P - 1 - c, Q - 1 - a, R - 2 - b		(B) P - 1 - b, Q - 2 - c, R - 1 - a
(C) P - 2 - b, Q - 1 - a, R - 1 - c		(D) P - 1 - c, Q - 1 - b, R - 2 - a

Answer (B)

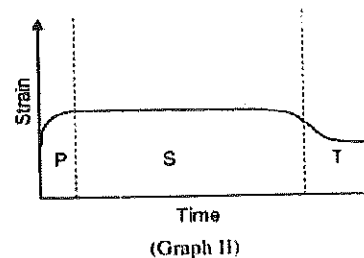
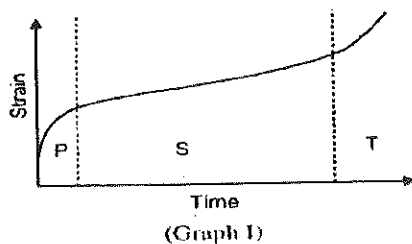
Q39. Match the following:

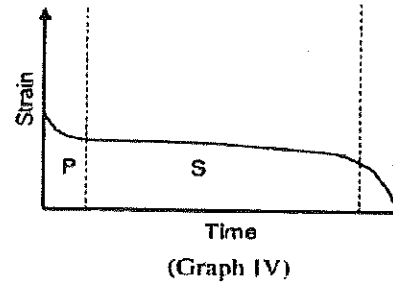
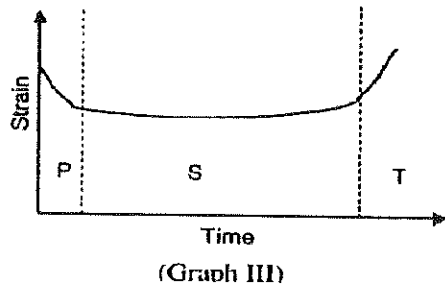
Stoping method	Advance of stoping face
P. Shrinkage stoping	1. Sideward vertical slices
Q. Rill stoping	2. Upward horizontal slice
R. Blasthole stoping	3. Downward horizontal slice
S. top slicing	4. Sideward inclined slices

- (A) P-3, Q-1, R-4, S-2 (B) P-2, Q-3, R-1, S-4
 (C) P-2, Q-4, R-1, S-3 (D) P-4, Q-3, R-2, S-1

Answer (C)

Q 40. Which one of the following graphs typically represents the standard strain-time creep behavior of an isotropic rock material under constant temperature? P, S and T in the figure refer to primary creep, secondary creep and tertiary creep respectively.





(A) Graph-I

(B) Graph-II

(C) Graph-III

(D) Graph-IV

Answer (A)

Q 41. The following data represent the number of workers suffering from pneumokoniosis in 10 coal mines.

Mine	I	II	III	IV	V	VI	VII	VIII	IX	X
Number	10	16	14	15	14	12	17	13	15	12

The number of mines falling above the 50th percentile in terms of the number of workers

Suffering from pneumokoniosis is

(A) 2

(B) 3

(C) 4

(D) 5

Answer (D)

Solution .

Mine (M)	Number(N)	Cumulative Number (CN)
I	10	10
II	16	26
III	14	40
IV	15	55
V	14	69
VI	12	81
VII	17	98

VIII	13	111
IX	15	126
X	12	138

Now, for 50th percentile = $\frac{138}{2} = 69$

So, number of mines falling above the 50th percentile = 5.

- Q 42. Cause –wise data for injuries in an underground coal mine for a five year period is given below:

Cause of Injury	Number of Injuries
Fall of roof	27
Fall of person	22
Rope haulage	17
Explosive	5
Other causes	4

The cumulative probability of injury due to fall of roof and fall of person is

- (A) 0.65 (B) 0.50 (C) 0.36 (D) 0.29

Answer (A)

Solution .

Cumulative probability of injury rate due to roof fall and fall of person is –

$$= \frac{27 + 22}{27 + 22 + 17 + 5 + 4}$$

$$= 0.65$$

- Q 43. Consider the following linear programming problem:

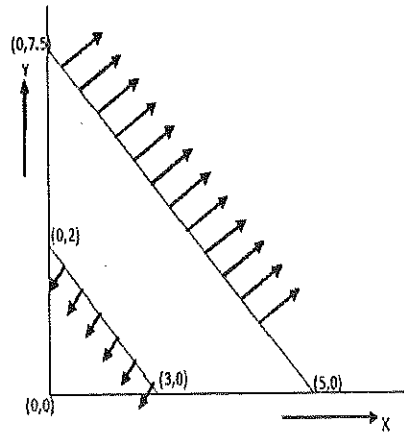
Maximize $z = 3x + 2y$
 Subject to $3x + 2y \geq 15$
 $2x + 3y \leq 6$
 $x \geq 0, y \geq 0$

The above linear programming problem has

- (A) unique optimal solution (B) multiple optimal solutions
 (C) unbounded solution (D) infeasible solution

Answer (D)

Solution .



∴ given linear programming has infeasible solution.

Infeasible solution- In linear programming, there is no common shaded area.

- Q 44.** A mine workshop has 4 lathe machines and 4 tasks for completion. Each of the machines can be perform each of the 4 tasks. Each task can be assigned to one and only one machine. Estimate cost in rupees to complete each task is given in the matrix below.

		Machine			
		M_1	M_2	M_3	M_4
Task	T_1	61	92	52	72
	T_2	42	49	69	85
	T_3	47	59	80	71
	T_4	65	70	68	72

The total optimum cost in Rupees for assigning the tasks to the machines is
(A) 210 (B) 215 (C) 220 (D) 286

Answer (C)

Solution .

	Machine			
	M_1	M_2	M_3	M_4
T_1	9	40	0	20
T_2	0	7	27	43
T_3	0	12	33	24
T_4	0	5	3	7

	Machine			
	M_1	M_2	M_3	M_4
T_1	9	35	0	13
T_2	0	2	27	36
T_3	0	7	33	17
T_4	0	0	3	0

	Machine			
	M_1	M_2	M_3	M_4
T_1	11	35	0	13
T_2	0	0	25	34
T_3	0	5	31	15
T_4	2	0	3	0

		Machine			
		M_1	M_2	M_2	M_4
Task	T_1	16	35	0	13
	T_2	5	0	25	34
	T_3	0	0	26	15
	T_4	7	0	3	0

Hence, Total optimum cost is = 52 + 49 + 47 + 72

= 220 rupees

- Q 45. A 1100 V, 3 Φ power supply system of a mine draws a load of 185 kW. The ammeter reading shows 115 a. The power factor of the system is
 (A) 0.84 (B) 0.73 (C) 0.64 (D) 0.48

Answer (A)

Solution.

We have,

$$\text{Power} = \sqrt{3} \times V_L \times I_L \times \cos \phi$$

$$185 \times 10^3 = \sqrt{3} \times 1100 \times 115 \times \cos \phi$$

$$\therefore \cos \phi = \frac{185 \times 10^3}{\sqrt{3} \times 1100 \times 115}$$

$$\therefore \cos \phi = 0.84$$

- Q 46. Two belt conveyors load a ground bunker, each at a rate of 400 tph, which is initially filled with 10000 t of coal. Coal is discharged from the bottom of the ground bunker onto a belt conveyor at a rate of 1200 tph. The time elapsed in hours before the bottom conveyor starts to operate below its rated capacity is

(A) 6.5

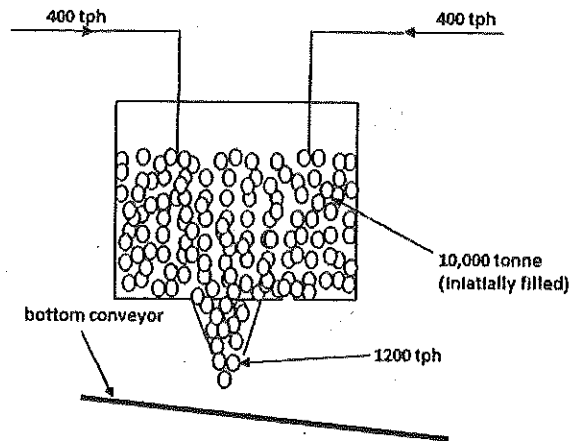
(B) 8.5

(C) 12.5

(D) 25.0

Answer (D)

Solution .



Here , 400 tonne of coal is produced from initially filled coal in the bunker to maintain discharge rate 1200 tph on bottom belt conveyor.

So, the time(T) till belt conveyor works at its rated capacity:-

$$T = \frac{10000}{400} = 25 \text{ hours}$$

Q 47. The case flow table of manganese mine for a particular year is shown be

Item	Amount (Rs. in lakhs)
Revenue	900
Cost (other than depreciation)	300
Depreciation	100
Profit before tax	500

If the corporate tax is 50% of the Profit before tax, the operating cash inflow in lakhs of Rupees is

(A) 400

(B) 350

(C) 250

(D) 200

Answer (C)

Solution .

Note - corporate tax is imposed on net profit.

Item	amount (Rs. in lakhs)
Revenue	900
Cost (other than depreciation)	300
Depreciation	100
Profit before tax	500

$$\therefore \text{corporate tax} = \frac{50}{100} \times 500 = 250 \text{ lakhs}$$

$$\therefore \text{operating cash flow} = 500 - 250 = 250 \text{ lakhs}$$

Q 48. In an area within a surface mine, under static condition the following gases are found:

NO_2 , CO_2 , O_3 and SO_2 . Assuming no diffusion, reaction and bonding of the gases, the concentration of the gases from bottom upwards will be in the order of

(A) NO_2 , CO_2 , O_3 and O_2

(B) SO_2 , NO_2 , CO_2 and O_3

(C) SO_2 , O_3 , NO_2 and CO_2

(D) NO_2 , CO_2 , SO_2 and O_3

Answer (C)

Q 49. In a mine site, the cost of shaft sinking in lakhs of Rupees is given as $2.64D + 34.8$, where D

is the shaft depth in m. In the same site, the corresponding cost of driving an incline is

$0.96L$, where L is the length of the incline in m. Assuming L by D ratio is 3.0, the depth in m

beyond which the shaft, sinking becomes more economical is

(A) 43

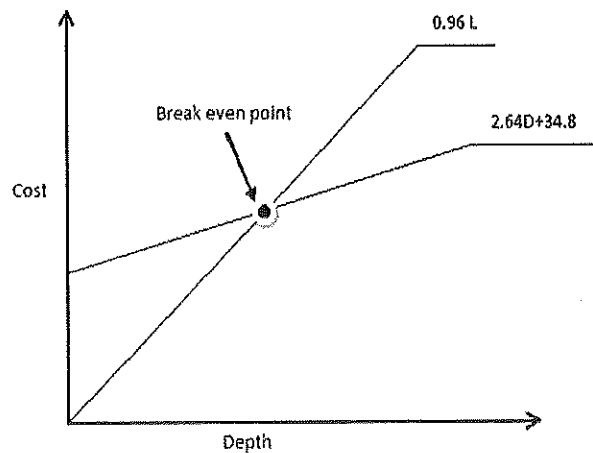
(B) 48

(C) 145

(D) 155

Answer (C)

Solution .



$$\frac{L}{D} = 3$$

At breakeven point,

$$C_S = C_I$$

$$2.64D + 34.8 = 0.96L$$

$$\therefore D = \frac{34.8}{0.24}$$

$$\therefore D = 145 \text{ m}$$

Q 50 .

Match the following:

Seam characteristics

P. 12 m thick flat seam

Q. 7 m thick seam at 65° inclination

R. 3 m thick flat seam

S. 7 m thick seam at 25° inclination

(A) P - 4, Q - 3, R - 2, S - 1

(C) P - 2, Q - 3, R - 4, S - 1

Coal mining method

1. Mechanized longwall

2. Descending shield

3. Mechanized integral caving

4. Jankowice

(B) P - 3, Q - 4, R - 1, S - 2

(D) P - 3, Q - 2, R - 1, S - 4

Common data for questions 51 and 52 :

Workmen arrive at a mine workshop to receive tools for maintenance . The inter-arrival

time of workmen at service counter is exponentially distributed with an average time of

10 min. the service time at the counter is also distributed exponentially with a mean time

of 6 min.

Q51. Probability that there is a queue (more than one workman) at the service counter is

(A) 0.24 (B) 0.36 (C) 0.40 (D) 0.60

Answer (D)

Solution .

$$\text{Arrival time} = \frac{1}{\gamma} = 10 \text{ min.}$$

$$\therefore \text{Arrival rate} = \gamma = \frac{1}{10} \text{ per min.}$$

$$\text{And service time} = \frac{1}{\mu} = 6 \text{ min.}$$

$$\therefore \text{Service rate} = \mu = \frac{1}{6} \text{ per min.}$$

Therefore ,

Probability that there is a queue (more than one workmen) at the service counter = traffic intensity or utilization ratio (ρ) :-

$$\rho = \frac{\gamma}{\mu} = \frac{\frac{1}{10}}{\frac{1}{6}} = 0.60$$

Q52. Average time spent by a workman waiting for this turn to be served in min is

(A) 9 (B) 12 (C) 15 (D) 18

Answer (C)

Solution .

Average time spent (T) by a workmen waiting for his turn to be served = $\frac{1}{\mu - \gamma}$

$$T = \frac{1}{\frac{1}{6} - \frac{1}{10}} = 15 \text{ minutes}$$

Common Data for Questions 53 and 54:

A tacheometer is set up at a station 'B'. The RL of the station B is 150 m and above the MSL

by holding staff vertically at a station 'A', the following readings are taken.

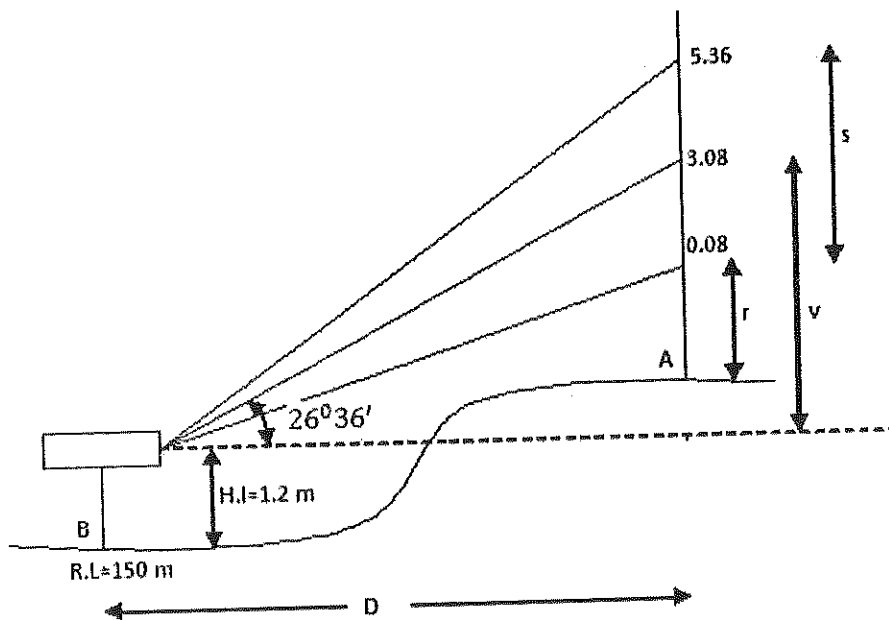
vertical angle	staff readings(m)		
$26^{\circ}36'$	Lower	middle	Upper
	0.08	3.08	5.36

The multiplying factor and additive constant of the instrument are 100 and 1.9 m respectively

- Q53. The horizontal distance between the stations A and B in m is
 (A) 364.6 (B) 366.3 (C) 409.4 (D) 457.6

Answer (B)

Solution .



Multiplying factor = 100, and $S = 5.36 - 0.80 = 4.56\text{ m}$

Additive constant = 1.9 m

We have,

$$D = KS (\cos \theta)^2 + (f + d) \cos \theta$$

$$D = 100 \times 4.56 \times (\cos 26^{\circ}36')^2 + 1.9 \times (\cos 26^{\circ}36')$$

$$\therefore D = 366.298\text{ m}$$

- Q 54.** If the height of the instrument is 1.2 m, the RL of the station 'A' above the MSL is m is
 (A) 337.6 (B) 334.5 (C) 331.5 (D) 330.3

Answer (C)

Solution .

We have,

$$V = KS \sin \theta \cos \theta + (f + d) \sin \theta$$

$$V = 100 \times 4.56 \times (\sin 26^\circ 36') \times (\cos 26^\circ 36') + 1.9 \times (\sin 26^\circ 36')$$

$$\therefore V = 183.45 \text{ m}$$

$$\therefore \text{R.L of station A above M.S.L} = 150 + 1.2 + 183.45 - 3.08 = 331.5 \text{ m}$$

Common Data for Questions 55 and 56:

A turbine pump of efficiency 70% discharges water at the rate of 2100 L/min at a total head of 100 m.

- Q55.** If the pump is run by a motor of efficiency 90%, the input power required for the motor in kW is
 (A) 22.49 (B) 34.31 (C) 44.11 (D) 54.50

Answer (D)

Solution .

$$\text{Input power (P) to motor} = \frac{P \cdot Q}{\text{efficiency}} \times 10^{-3} \text{ kW}$$

$$P = \rho \cdot g \cdot h = 1000 \times 100 \times 9.81 = 981000 \text{ Pa}$$

$$Q = 2100 \frac{\text{L}}{\text{min}} = 0.035 \text{ m}^3/\text{sec}$$

$$\text{Now, input power (P) to motor} = \frac{981000 \times 0.035 \times 10^{-3}}{0.70 \times 0.90} \text{ kW}$$

$$P = 54.5 \text{ kW}$$

Q56. If the velocity of water in suction and delivery pipes of the pump are 1.8 m/s and 2.5 m/s

respectively, the diameter of suction and delivery pipes in cm are

- (A) 15.73 and 13.35 (B) 7.86 and 6.67
(C) 5.78 and 6.02 (D) 4.97, 4.22

Answer (A)

Solution .

We have,

$$Q = A_1 V_1 = A_2 V_2$$

$$Q = 2100 \frac{L}{min} = 0.035 \text{ m}^3/sec$$

$$V_1 = 1.8 \text{ m/sec}, V_2 = 2.5 \text{ m/sec}$$

$$\text{Now, } Q = A_1 V_1 \text{ [in the suction side]}$$

$$\therefore 0.035 = 1.8 \times A_1$$

$$\text{Or } A_1 = \frac{0.035}{1.8}$$

$$\text{Or } \frac{\pi}{4} \times d_1^2 = \frac{0.035}{1.8}$$

$$\therefore d_1 = 0.1573 \text{ m} = 15.73 \text{ cm}$$

And in the delivery side-

$$Q = A_2 V_2$$

$$0.035 = A_2 \times 2.5$$

$$\therefore d_2 = 0.1335 \text{ m} = 13.35 \text{ cm}$$

Statement for Linked Answer Questions 57 and 58:

A fan running at a speed of 280 rpm circulates $105 \text{ m}^3/s$ of air in a mine.

Q 57. If the power input to the motor for driving the fan is recorded to be 75 kW, with the combined efficiency of fan and motor at 70 %, the fan pressure in Pa is

- (A) 50 (B) 350 (C) 500 (D) 650

Answer (C)

Solution .

We have,

$$\text{Input power (P) to motor} = \frac{P \cdot Q}{\text{efficiency}} \times 10^{-3} \text{ kW}$$

$$75 = \frac{P \times 105}{0.7} \times 10^{-3}$$

$$\therefore P = 500 \text{ Pa}$$

Q58. If the fan pressure is to be increased by 200 Pa by changing the fan speed, the fan speed in rpm will become

(A) 768 (B) 549 (C) 392 (D) 332

Answer (D)

Solution .

We have,

$$\text{pressure} \propto (\text{speed})^2$$

$$\text{Or } \frac{P_1}{P_2} = \left(\frac{N_1}{N_2} \right)^2$$

Given that,

$$P_1 = 500 \text{ pa}$$

$$P_2 = 500 + 200 = 700 \text{ pa}$$

$$N_1 = 280 \text{ rpm}$$

$$\therefore \frac{500}{700} = \left(\frac{280}{N_2} \right)^2$$

$$\therefore N_2 = 332 \text{ rpm}$$

Statement for Linked Answer Questions 59 and 60:

A surface mine blast design has 9 holes in a row, each of 8 m length and 200 mm diameter.

the spacing and burden are 6 and 5 m respectively. The length of subgrade drilling is 1 m and

the density of in-situ rock is 2.43 t/m^3 .

Q59. Assuming no back break, the output per blast in t is

- (A) 4593 (B) 5905 (C) 6124 (D) 6299

Answer (A)

Solution .

We have,

$$\begin{aligned}\text{output per blast} &= \text{No. of holes} \times \text{spacing} \times \text{burden} \times \text{density} \\ &\quad \times (\text{length of holes} - \text{subgrade length}) \\ &= 9 \times 6 \times 5 \times (8 - 1) \times 2.43 = 4593 \text{ tonnes}\end{aligned}$$

Q60. Considering an explosive density of 0.9 t/m^3 and stemming length of 2 m, the powder factor from the blast in t/kg

- (A) 4.00 (B) 4.12 (D) 3.01 (C) 3.86

Answer (D)

Solution .

$$\begin{aligned}\text{Explosive required per round of blasting} &= \text{cross sectional area of hole} \times \text{No. of hole} \\ &\quad \times (\text{length of hole} - \text{stemming height}) \times \text{explosive density} \\ &= \frac{\pi}{4} \times 0.2^2 \times 9 \times (8 - 2) \times 0.9 \times 1000 \\ &= 1526.04 \text{ kg}\end{aligned}$$

$$\begin{aligned}\therefore \text{Powder Factor} &= \frac{\text{output per blast (tonne)}}{\text{explosive required (kg)}} \\ &= \frac{4592}{1526.04} \\ &= 3.01 \text{ t/kg}\end{aligned}$$